Alec Truax

alectruax.github.io
alec.truax@colorado.edu
(+1) 406-439-2498

Department of Economics
256 UCB
Boulder, CO 80309

Education

University of Colorado, Boulder *Expected, May 2027*
Ph.D. in Economics

University of Colorado, Boulder *May 2023*
M.A. in Economics

Montana State University, Bozeman *May 2021*
B.S. in Economics
Minors in Mathematics & Statistics

Research Interests

Energy Economics, Environmental and Resource Economics, International Trade

Job Market Paper

Alec Truax. "Retail Electricity Prices and Renewable Generation in the United States"
Abstract: *While zero-marginal-cost renewable energy has reduced wholesale electricity prices, its effect on retail electricity prices remains uncertain. A common concern is that costly investments in renewable integration, such as transmission expansion, generate system-level costs that may offset or even exceed savings from lower generation costs. This paper tests this conjecture empirically using annual sales data from the EIA 861 and electricity generation data from the EIA 923 for the majority of the United States between 2004 and 2023. To address the inherent simultaneity between prices and energy generation, I leverage spatial variation in wind and solar potential and time-varying levelized costs of wind and solar energy to construct a Bartik-style instrument to predict renewable penetration. I find that a one percentage point increase in renewable penetration at the balancing authority level reduces real utility level yearly average retail prices by 0.3%, implying a 3.5% decrease in real retail prices at the mean change in renewable penetration. Crucially, OLS estimates that do not account for endogeneity are positively biased and insignificant. Even under alternate instruments and sample restrictions, the estimated effect is tightly bounded between - 0.7% and 0.7%, suggesting that increased renewable penetration has had, at most, a modest impact on energy affordability. Back of the envelope calculations suggest that system costs offset roughly two-thirds of the marginal generation cost savings estimated in the literature.*

Working Papers

Jeong-won Ko, Sergey Nigai, and Alec Truax. "Sectoral Transport Mode Elasticities and International Trade", *R&R at Journal of International Economics*
Alec Truax. "The Anatomy of Environmental Provisions in Trade Agreements"

Works in Progress

Stephanie Weber, Paige Weber, and Alec Truax. "The Effect of Electricity Interchange on Localized Marginal Prices: an IV Approach"

Conference Presentations

2027 – "*Retail Electricity Prices and Renewable Generation in the United States*"
Allied Social Science Associations Meeting — *Forthcoming - TPUG sessions*

2026 – "*Retail Electricity Prices and Renewable Generation in the United States*"
Eastern Economics Association Annual Conference — *AERE sessions*
7th World Congress of Environmental and Resource Economists — *Session*

Teaching

Instructor of Record
Department of Economics, University of Colorado Boulder

- Natural Resource Economics — Spring 2025
- Principles of Microeconomics — Fall 2023, Fall 2024

Leeds School of Business, University of Colorado Boulder

- Principles of Microeconomics *(EXCEL Scholars Program)* — Summer 2026

Teaching Assistant
Department of Economics, University of Colorado Boulder

- Principles of Microeconomics — Fall 2021, Fall 2022, Spring 2023, Spring 2024, Fall 2026
- Principles of Macroeconomics — Spring 2022, Spring 2026

Department of Agricultural Economics and Economics, Montana State University

- Economic Policy Evaluation — Spring 2021
- Intermediate Microeconomics with Calculus — Fall 2019 - Spring 2021
- Introduction to Microeconomics — Fall 2020 - Spring 2021

Department of Mathematical Sciences, Montana State University

- Introduction to Statistics — Fall 2019

Tutorship
Veteran's Services Center, Montana State University

- Statistics and Economics — January 2019 - May 2021

Honors and Awards

Stanford Calderwood Student Teaching Award — 2025, 2026
SSCI Fellowship — 2025
Reuben A. Zubrow Fellowship in Economics — 2023, 2024
Prize in Econometrics — 2023
MSU Award for Excellence — 2021
Outstanding Graduating Senior in Economics Department of Agricultural Economics and Economics — 2021
College of Letters and Science General Scholarship — 2020, 2021
IRAEA Undergraduate Research Scholarship — 2020

Professional Activities and Affiliations

Referee: *Resource and Energy Economics*

Abstract Reviewer: *AERE @ Southern Economic Conference* — 2026

Organizer: *Graduate Student Brownbag at CU Boulder* 2025-2027

Affiliations:
Association of Environmental and Resource Economists, Western Economic Association, Eastern Economic Association, Transportation and Public Utilities Group, Phi Kappa Phi

Skills

Software: Stata, Python, LaTeX, Git, R, and Matlab
Languages: English *(native)*

References

Daniel Kaffine *Co-Chair*
Professor, Department of Economics
University of Colorado, Boulder
em: daniel.kaffine@colorado.edu
phone: 303.492.6652

Sergey Nigai *Co-Chair*
Associate Professor, Department of Economics
University of Colorado, Boulder
em: sergey.nigai@colorado.edu
phone: 303.492.0748

Wolfgang Keller *Committee Member*
Professor, Department of Economics
University of Colorado, Boulder
em: wolfgang.keller@colorado.edu
phone: 303.735.5507

Stephanie Weber *Committee Member*
Assistant Professor, Department of Economics
University of Colorado, Boulder
em: stephanie.m.weber@colorado.edu
phone: 303.735.5500